

ENSEMBLE AND PERSONALIZED TRANSFORMER MODELS FOR SUBJECT IDENTIFICATION AND RELAPSE DETECTION IN E-PREVENTION CHALLENGE

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ABSTRACT

In this short paper, we present the devised solutions for the subject identification and relapse detection tasks, which are part of the e-Prevention Challenge hosted at the ICASSP 2023 conference [1] [2] [3]. We specifically design an ensemble scheme of six models - five transformer-based ones and a CNN model - for the identification of subjects from wearable devices, while a personalized - one for each subject - scheme is used for relapse detection in psychotic disorder. Our final submitted solutions yield top performance on both tracks of the challenge: we ranked 2nd on the subject identification task (with an accuracy of 93.85%) and 1st on the relapse detection task (with a ROC-AUC and PR-AUC of about 0.65). Code and details are available at <https://github.com/perceivelab/e-prevention-icassp-2023>.

1. INTRODUCTION

This paper describes the submissions by the PeRCeiVe Lab team for task 1 and 2 of the e-Prevention Signal Processing Grand Challenge. The objective of the challenge is the use of biosignals from wearable devices for subject identification and relapse detection in the psychotic spectrum. The provided data includes daily continuous accelerometer and gyroscope measurements, together with heart rates and RR intervals. Sleeping information and step data are also included. For the subject identification task, we designed an ensemble model consisting of a convolutional neural network and five different transformer-based models to distinguish between 46 subjects. The final prediction is obtained by summing the prediction scores of the ensemble models. For the relapse detection task, we instead designed a different autoencoder (based on either CNN or on transformers) for each subject to learn the distribution of non-relapse data. The learned latent distribution was then used to identify anomalies. During validation and test, we use the cumulative distribution function (CDF) of normal data reconstruction error to compute anomaly probabilities.

2. METHODS

2.1. Track 1: Subject identification

All experiments are conducted on the data collected within the e-Prevention project [1] that includes daily continuous recordings (linear and angular accelerations, heart rates, RR intervals, sleeping information and step data - number of steps, distance and calories) from smartwatches of 46 subjects, with a mean of 2.5 months of recordings per subject.

The e-Prevention dataset is already split into train, validation and test sets and labels are provided only for training and validation sets. We normalize the accelerometer, gyroscope, heart rate and RR intervals values in $[0, 1]$. The binary sleeping information is used without any change. From the step data, we derive three sequences, using the provided timestamps, namely, the number of steps per time unit, the velocity expressed in m/s and the calories per time unit.

Data is augmented with a sliding window approach using non-overlapping windows with different widths (1.5 and 3 hours are used). The input sequence has a $T \times F$ shape, where T is the time dimension (2160 timepoints for 3h windows, 1860 for 1.5h) and F is the number of features corresponding to 12 series data (3 accelerometer, 3 gyroscope, heart rate, RR intervals, sleeping activity and the three step series).

The subject identification model consists of an ensemble between a standard deep 1D CNN and five transformer architectures. The CNN model serves also as the embedding layer for all the transformer models. We use Time2Vec [4] as Positional Embedding and we prepend a $[CLS]$ token to the final sequence, which is fed into the encoder. The $[CLS]$ token of the last layer is used as input to a final classifier. The different transformer configurations, included in the ensemble, are obtained by modifying positional encoding, model's depth, encoding layers and attention heads.

For each model, the final parameters are chosen based on the validation set and inference prediction is performed by summing the prediction logits of each model in the ensemble. For training, only windows with 83% of valid data are considered. Invalid data were imputed with NN interpolation.

2.2. Track 2: Relapse detection

The data structure for relapse detection is similar to the previous task, but here only 10 subjects are present and recordings are divided into *relapse* and *non-relapse* data on a daily basis. We use both *raw* and *aggregated* data in our approach. *Raw* data is processed as for the previous task. In the *aggregated* approach, we extract 10 features (mean norm of linear and radial accelerations, mean heart rate and RR interval, major axis of the Poincare ellipse and the normalized low and high frequency powers of the Lomb-Scargle periodogram from the NNI series, daily sinusoidal encoding, percentage of valid samples) from 5-minute slices of the original data. Data is standardized using per-patient mean and standard deviation computed on the training set.

We pose the relapse detection task as an anomaly detection one in a personalized scheme, i.e., we designed a different model (referred in the following to as *patient model*) for each subject. Three different architectures are employed: a CNN-based autoencoder, an autoencoder for time series¹ and a Transformer Encoder-Decoder. Each *patient model* is trained separately on both *raw* and *aggregated* data and model selection is carried out using the best accuracy on the validation set. During training only normal samples are considered, and the models are trained to reconstruct the inputs using MSE loss for 100 epochs. After training, we compute the Cumulative Distribution Function $CDF(x) = p(X \leq x)$ of per-channel reconstruction error. During inference, we compute the reconstruction error e and we use $CDF(e)$ as an anomaly score. Since every day has different windows, the evaluation is carried out using the median anomaly score over all available windows.

Architectural details, training procedures and hyperparameters for the employed models are available at the github link provided in the abstract.

3. RESULTS

The proposed models were evaluated on the test sets by the challenge organizers, respectively, using the raw accuracy score for Track 1 and the harmonic mean of ROC-AUC and PR-AUC for Track 2. The results of our solutions were compared with those obtained by the other challenge participants and shown, respectively, on Table 1 and Tab. 2. Obtained performance clearly indicates the effectiveness of the designed models as they ranked, respectively, 2nd position for subject identification task and 1st position on relapse detection task.

4. ACKNOWLEDGEMENTS

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¹<https://github.com/EUROVOLC-ML/VolUnD-Toolkit>

Table 1. Performance comparison among participant teams on the provided test set for subject identification. We report the raw test accuracy for each team. Our score ranks 2nd.

Team	Accuracy
SRCB-LUL	95.00%
PeRCeiVe	93.85%
Al_Bezzie	91.36%
SAILers	82.15%
unipi_cmb1	75.43%
ADCADD	3.83%
ID-EPRE2	2.88%
eHust	2.88%
CogBCI	2.88%
uoi	2.68%
NWPU	2.3%

Table 2. Performance comparison among participant teams on the provided test set for relapse detection. We report the ROC-AUC, the PR-AUC and the harmonic mean of the previous scores, for each team. Our score ranks 1st.

Team	ROC-AUC	PR-AUC	Total
PeRCeiVe	0.6469	0.6509	0.6489
Emotion	0.6072	0.6347	0.6209
SAILers	0.5839	0.6263	0.6051
SmartBCI	0.5435	0.5863	0.5604
YDH@HEU	0.5215	0.5587	0.5401
GIPS@HEU	0.5117	0.5480	0.5229

5. REFERENCES

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